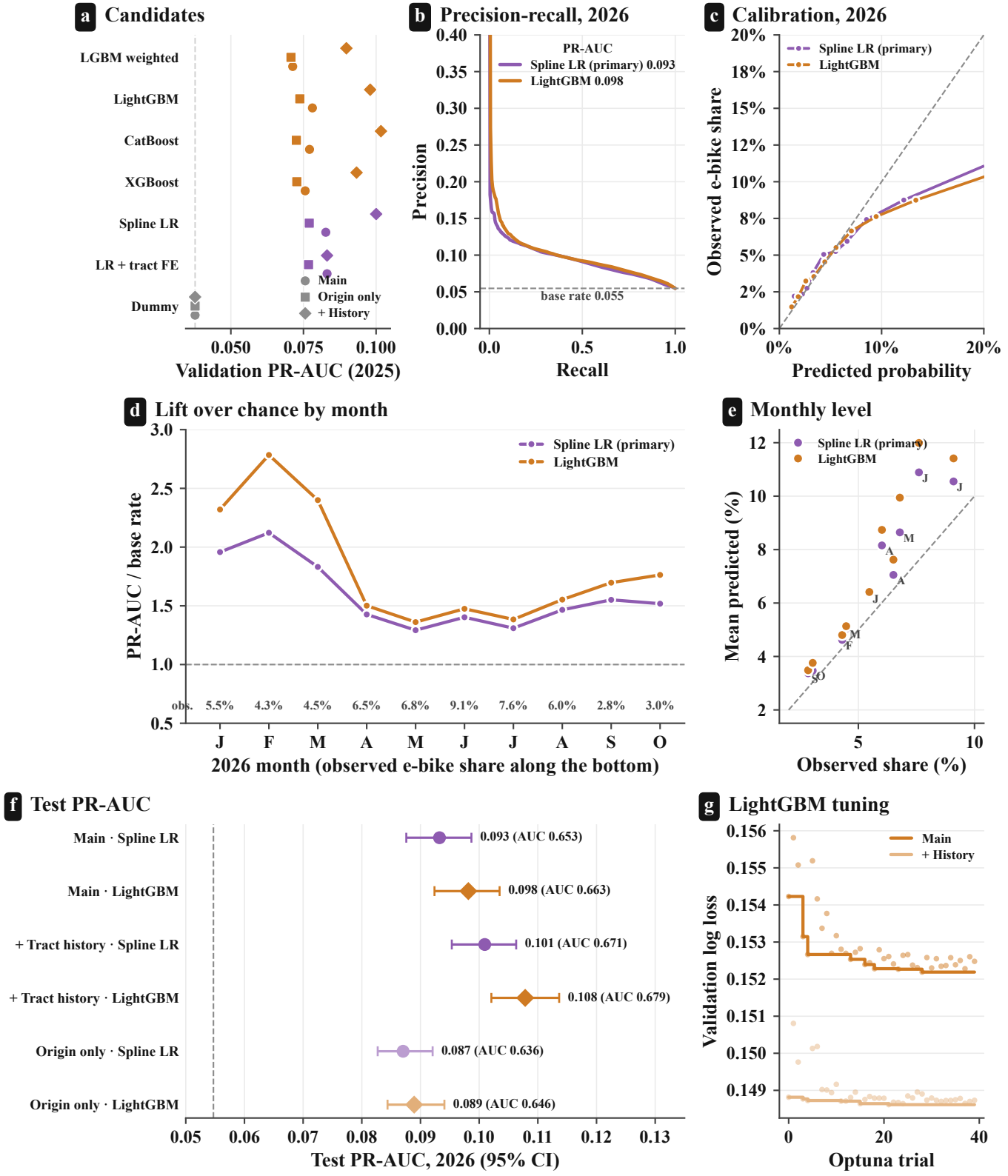
a Candidates
LGBM weighted
LightGBM
CatBoost
XGBoost
Spline LR
LR + tract FE
Dummy
Main
Origin only
+ History
0.050
0.075
0.100
Validation PR-AUC (2025)
b Precision-recall, 2026
PR-AUC
Spline LR (primary) 0.093
LightGBM 0.098
base rate 0.055
Precision
Recall
c Calibration, 2026
Spline LR (primary)
LightGBM
Observed e-bike share
Predicted probability
d Lift over chance by month
Spline LR (primary)
LightGBM
PR-AUC / base rate
obs. 5.5% 4.3% 4.5% 6.5% 6.8% 9.1% 7.6% 6.0% 2.8% 3.0%
J F M A M J J A S O
2026 month (observed e-bike share along the bottom)
e Monthly level
Spline LR (primary)
LightGBM
Mean predicted (%)
Observed share (%)
f Test PR-AUC
Main · Spline LR
0.093 (AUC 0.653)
Main · LightGBM
0.098 (AUC 0.663)
+ Tract history · Spline LR
0.101 (AUC 0.671)
+ Tract history · LightGBM
0.108 (AUC 0.679)
Origin only · Spline LR
0.087 (AUC 0.636)
Origin only · LightGBM
0.089 (AUC 0.646)
Test PR-AUC, 2026 (95% CI)
g LightGBM tuning
Main
+ History
Validation log loss
Optuna trial